

Likelihood-Ranked Synthetic Selection in Self-Consuming Diffusion: Minority Extinction Ordering at Moderate Imbalance

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Abstract

Self-consuming training loops—retraining generative models on their own outputs—can induce *model collapse*, including loss of rare modes. We study unlabeled selection of synthetic samples by a denoising likelihood proxy in self-consuming DDPM training on imbalanced mixtures, with emphasis on **moderate** imbalance. Define the minority-mass ratio $r = m^{(1)}/m^{(0)}$. Results are **stratified by ρ** . At $\rho=5$ ($n=5$), generation 1 minority mass obeys bottom $>$ rand $>$ top on all seeds with large paired effects vs. random- k (top–rand $d_z=-1.96$; bottom–rand $d_z=+1.65$); we treat this as a *directional* finding because exact Wilcoxon $p=0.0625$ is the discrete minimum at $n=5$ and cannot reach $p<0.05$. At $\rho=10$ ($n=3$), the order fails (0/3) and contrasts are near null, consistent with a minority-mass floor. Our strongest powered claim is that top- k does *not* improve sliced W_2 at generation 1 ($\Delta W_2 > 0$ on 8/8 cells; exact $p=0.0078$; Holm $p=0.039$). Directional transfers (8D GMM, Digits-PCA; $n=3$) match the $\rho=5$ order; a Digits-CVAE check was inconclusive. Code: <https://github.com/muhammadalamzeb/likelihood-rank-self-consuming-diffusion>.

1 Introduction

As synthetic data proliferates, generative models are increasingly trained on mixtures of real and model-generated samples. Iterated self-consumption can cause *model collapse* [Shumailov et al., 2023, Alemohammad et al., 2023, Gerstgrasser et al., 2024]. Mitigations include accumulating real data [Gerstgrasser et al., 2024] and filtering synthetics [Feng et al., 2024, Cai et al., 2025]. When the only score is the generator’s own likelihood, how should one select synthetics? We isolate an *interventional* selection policy at matched synthetic count and matched real-mix α , with mode-exact metrics.

What is new. A controlled top- k / random- k / bottom- k contrast under matched budgets, with mode-exact minority retention and an explicit falsification of “top- k looks better on W_2 ”—not a new named training algorithm or ImageNet-scale SOTA method.

Contributions. (1) Matched-budget DDPM protocol with exact mode masses. (2) Directional likelihood-rank ordering of minority retention at moderate imbalance ($\rho=5$), **stratified by ρ** . (3) Powered evidence that top- k does not improve sliced W_2 vs. random- k at generation 1. (4) Directional 8D GMM and Digits-PCA checks (Digits-CVAE inconclusive).

2 Related work

Shumailov et al. [2023], Alemohammad et al. [2023] document collapse and quality-biased sampling (*MAD*). Gerstgrasser et al. [2024] show accumulating real+synthetic data can avoid collapse. Feng et al. [2024] study task verification of labeled synth; Cai et al. [2025] (*LSF*) filter via latent probes. Qiao et al. [2026] prove that selection against a *biased local* real-data reference can prune global tail modes and accelerate collapse in siloed settings; their remedy uses collaborative Wasserstein proxy references. We study a complementary failure mode: ranking synthetics by the *generator’s own* denoising-likelihood proxy, with a matched-budget top- k / random- k / bottom- k contrast and minority-mass retention as the primary diversity metric. Inference-time temperature methods [Xu et al., 2025, Li et al., 2026] reshape sampling diversity; we intervene on training-set selection.

3 Method

3.1 Self-consuming loop

Train DDPM $f^{(g)}$ on $\mathcal{D}^{(g)}$, sample synthetics, mix fixed real buffer (fraction α) with selected synthetics.

3.2 Likelihood proxy and policies

Score $s(x)$: mean denoising MSE over a t -grid (lower = higher likelihood).

- **top- k / bottom- k / random- k :** pool of size $k = \max(n_{\text{syn}}, \lfloor k_{\text{frac}} N \rfloor)$ by lowest / highest / uniform scores; then draw n_{syn} and mix with α real.
- **mix:** same α , but synthetic portion is a uniform subsample of all N synthetics (no rank filter).
- **replace:** pure synthetic.

Thus mix $\neq$ random- k .

3.3 Metrics

Nearest-mean mode masses; minority-mean m ; ratio $r = m^{(1)}/m^{(0)}$ (“retention,” allowed > 1). Sliced W_2 : mean squared 1D Wasserstein over **24** projections; ≤ 400 synth and ≤ 400 held-out real points. Hypothesis tests use the **exact** Wilcoxon signed-rank distribution over sign assignments (enumeration; midranks for magnitude ties), not only a normal approximation.

4 Experiments

4.1 Primary design ($n=8$ cells)

$K=4$, $G=5$, $\alpha=0.5$, $k_{\text{frac}}=0.5$. Logs: `w2_fs.jsonl`.

Table 1: Primary cells. Seeds 3–4 only at $\rho=5$.

$\rho \backslash$ seed	0	1	2	3	4
5	✓	✓	✓	✓	✓
10	✓	✓	✓	—	—

Table 2: Per- ρ mean r and seed-wise counts (generation 1).

ρ	n	bottom	rand	top	Full order	top<rand / bot>rand
5	5	0.737	0.456	0.183	5/5	5/5 / 5/5
10	3	0.296	0.199	0.154	0/3	2/3 / 1/3

4.2 Retention: stratified primary analysis

Because $\rho=5$ and $\rho=10$ behave differently, we do **not** treat a single pooled Wilcoxon as the primary claim. Table 2 gives means and seed-wise order counts; Table 3 gives stratified paired contrasts.

At $\rho=5$, all five seeds agree in sign with large effects ($|d_z|>1.6$). Exact two-sided $p=0.0625$ is the smallest attainable at $n=5$ under the signed-rank null when all signs agree; the test *cannot* reject at $\alpha=0.05$ no matter how large the effect. We therefore report the $\rho=5$ ordering as **directional** / **suggestive** (unanimous signs + effect sizes + CIs excluding zero for the mean Δ), not as a claim that clears a conventional significance threshold. At $\rho=10$, minorities are already near floor at $g=0$: few minority synthetics remain to rank, so top/random/bottom pools become statistically similar and contrasts are consistent with null (a saturation / floor hypothesis; isolating equal-minority-count designs is left to future work). For completeness, a pooled $n=8$ exact Wilcoxon yields $p=0.039$ for both contrasts, but we treat that as secondary because it mixes heterogeneous strata.

4.3 Sliced W_2

Generation-wise tests are repeated measures on the same cells (serially correlated), so Holm is a conservative acknowledgment rather than an assumption of independence. Evidence against top- k improving W_2 is strongest at $g=1$.

4.4 Real-mix α ablation (illustrative; $n=2$)

At $\alpha=0.5$, seed 0 breaks the full order (bottom $m^{(1)}$ slightly below random) while seed 1 holds—hence 1/2 (means still rank bottom>rand>top). At $\alpha=0.25$, likewise 1/2; at $\alpha=0.75$, top vs. random reverses on average (0/2).

4.5 Transfer checks (directional; $n=3$)

Minimum two-sided exact Wilcoxon p at $n=3$ with all signs agreeing is 0.25; we report Cohen’s d_z against that bound and do not claim powered generalization.

- **8D GMM:** 0.55 ± 0.20 / 0.27 ± 0.06 / 0.08 ± 0.03 ; order 3/3; top–rand $d_z = -2.56$, bottom–rand $d_z = +1.28$ (exact $p=0.25$).
- **Digits-PCA:** 1.49 ± 0.11 / 1.11 ± 0.08 / 0.68 ± 0.17 ; order 3/3; top–rand $d_z = -1.72$, bottom–rand $d_z = +2.67$ (exact $p=0.25$).

Table 3: Stratified paired retention contrasts vs. random- k (exact Wilcoxon; bootstrap 95% CI of mean Δ).

ρ	Contrast	Mean Δ	d_z	95% CI	Exact p
5	top-rand	-0.273	-1.96	$[-0.37, -0.15]$	0.0625
5	bottom-rand	+0.281	+1.65	$[0.16, 0.43]$	0.0625
10	top-rand	-0.045	-0.28	$[-0.19, 0.13]$	0.75
10	bottom-rand	+0.097	+0.38	$[-0.08, 0.39]$	1.00

Primary retention by imbalance (stratified)

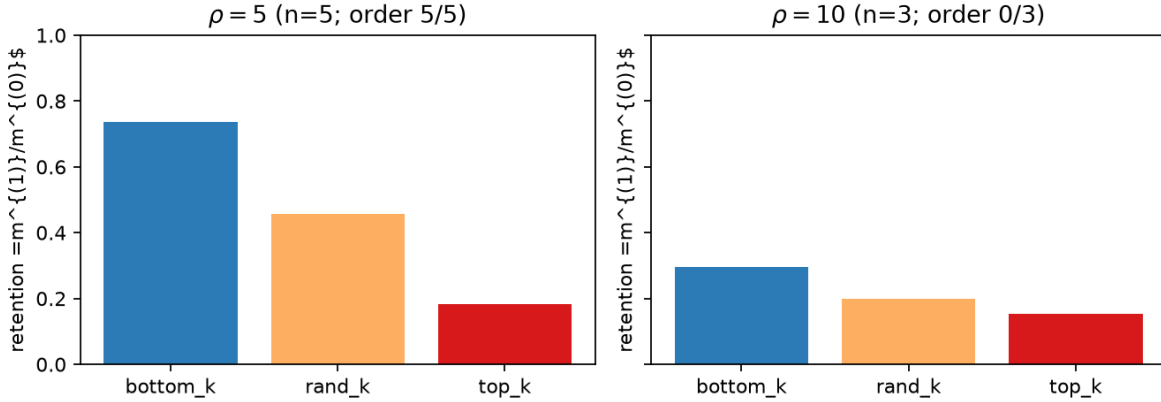


Figure 1: Primary retention r stratified by ρ (matches the primary claim). A pooled bar chart would mix the 5/5 and 0/3 regimes and is not used as the lead figure.

- **Digits-CVAE (inconclusive):** a pixel-space conditional VAE transfer did not reproduce a stable bottom>rand>top order under the same budget protocol; we treat Digits-PCA (mode masses in PCA space) as the supported image-adjacent check and do not claim CVAE replication.

4.6 Selection-pool size k_{frac} (illustrative; $n=2$)

Top- k remains worst on means; at 0.75, bottom vs. random flips (0/2 full order).

5 Analysis

Top- k retains high-likelihood (typically majority) synthetics, reducing minority inflow. The ordering is directionally clear when minorities remain measurable ($\rho=5$) and near noise at $\rho=10$, consistent with a floor on rankable minority mass. Sliced W_2 is a poor monitor of minority survival here: aggregate transport can look acceptable while rare-mode retention collapses.

6 Limitations

Toy GMMs / Digits-PCA; not ImageNet-scale. At $\rho=5$, exact Wilcoxon cannot beat $p=0.0625$ with $n=5$ all-agreeing signs—retention inference is directional (d_z , unanimous signs); the powered

Table 4: $\Delta W_2(\text{top} - \text{rand})$ by generation on all eight cells (positive = top worse). Exact Wilcoxon p ; Holm adjustment across the five (non-independent) generation-wise tests; bootstrap 95% CI of the mean.

g	Mean	Std	d_z	95% CI	$\#\{> 0\}$	Exact p (Holm)
1	+0.113	0.099	1.14	[0.05, 0.18]	8/8	0.0078 (0.039)
2	+0.061	0.060	1.00	[0.02, 0.10]	6/8	0.039 (0.094)
3	+0.039	0.035	1.11	[0.02, 0.06]	7/8	0.023 (0.094)
4	+0.027	0.032	0.83	[0.01, 0.05]	7/8	0.023 (0.094)
5	+0.024	0.033	0.73	[0.01, 0.05]	7/8	0.023 (0.094)

Table 5: Mean r by α (seeds $\{0, 1\}$, $\rho=5$, default $k_{\text{frac}}=0.5$). “Order holds” = bottom>rand>top on $g=1$ minority mass. The $\alpha=0.5$ row is the *same cell* as $k_{\text{frac}}=0.5$ in Table 6.

α	bottom	rand	top	Order holds
0.25	0.25	0.17	0.05	1/2
0.50	0.62	0.49	0.24	1/2
0.75	1.34	0.77	0.94	0/2

negative result is the W_2 contrast at $g=1$. α and k_{frac} ablations use two seeds; their default rows are the same experimental cell. Transfers: $n=3$ cannot beat exact $p=0.25$; Digits-CVAE was inconclusive. **Multiplicity:** Holm is applied within families (two $\rho=5$ retention contrasts; five W_2 generations). Additional stratified/exploratory tests are not multiplicity-corrected paper-wide; we treat them as descriptive under the small-sample framing. Contribution is a controlled empirical contrast complementary to MAD/LSF/Feng and to siloed-selection theory [Qiao et al., 2026].

Exploratory search and interpretation of p -values. The reported claim is the survivor of an earlier exploratory search over several small experimental stacks (archived under `docs/` and `experiments/` in the public repository), not a single pre-registered hypothesis tested once. Readers should interpret retention p -values as descriptive under that history; we mitigate overclaiming by stratifying on ρ , demoting pooled tests, labeling $\rho=5$ retention as directional, and treating the W_2 generation-1 result as the primary powered claim.

Note on double-blind review. The public GitHub URL reveals author identity. For double-blind venues, provide an anonymous code mirror; for arXiv, the named repository is appropriate.

7 Conclusion

At moderate imbalance ($\rho=5$), likelihood ranking yields a directional minority-retention order bottom > rand > top; the order is not seed-wise reliable at $\rho=10$ in our matrix, consistent with a minority-mass floor. Top- k does not improve sliced W_2 vs. random- k at generation 1. Preferring “most likely” synthetics is not a free lunch for minority survival when rare modes remain measurable.

Table 6: k_{frac} ablation (seeds $\{0, 1\}$, $\rho=5$, $\alpha=0.5$). $0.25 \equiv 0.5$ when n_{syn} binds. The $k_{\text{frac}}=0.5$ row duplicates Table 5’s $\alpha=0.5$ row.

k_{frac}	bottom	rand	top	Order holds
0.25	0.62	0.49	0.24	1/2
0.50	0.62	0.49	0.24	1/2
0.75	0.48	0.61	0.22	0/2

References

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A Reproducibility

`python scripts/reproduce_w2.py`; stats tables via `experiments/scripts/make_submission_stats.py`.
Code: <https://github.com/muhammadalamzeb/likelihood-rank-self-consuming-diffusion>.